

Long Li

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Employment

- 2024.02 - **Research Scientist**, Johann Radon Institute for Computational and Applied Mathematics, Austrian Academy of Sciences, Linz, Austria.
(Mentor: Prof. Mourad Sini)
- 2021.07 - **Postdoctoral Fellow**, Institute of Applied Mathematics, Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Beijing, China.
2024.01 (Mentor: Prof. Bo Zhang)

Education

- 2016.07 - **PhD in Mathematics**, School of Mathematical Sciences, Peking University, Beijing, China.
2021.06 (Supervisor: Prof. JianSheng Yang)
- 2012.09 - **Bachelor in Mathematics**, School of Mathematics and Statistics, Xidian University, Xi'an, China.
2016.07

Research Interests

- Wave propagation and resonance phenomena in complex and structured media
- Inverse Problems for partial differential equations and imaging

Publications

Preprints

- 12 Long Li and Mourad Sini, A time-domain pressure–interface model for gas bubble dynamics with surface tension: well-posedness, classical limits, and resonance branches, arXiv:2607.06064.
- 11 Long Li and Mourad Sini, High contrast transmission resonance for the Lamé system, arXiv:2601.10290, submitted to Journal of Functional Analysis.

Journal Articles

- 10 Long Li and Mourad Sini, High contrast transmission and Fabry-Pérot-type resonances, arxiv:2510.19096v1, accepted by *Communications in Partial Differential Equations*.
- 9 Long Li and Mourad Sini, Uniform resolvent estimates for subwavelength resonators: the Minnaert bubble case, *SIAM Journal on Mathematical Analysis* 58 (2026), 2635–2674.

- 8 Long Li and Mourad Sini, Uniform space and time behavior for acoustic resonators, *Multiscale Modeling & Simulation. A SIAM Interdisciplinary Journal* 24 (2026), 563-603.
- 7 Haiyang Liu, Long Li, Jiansheng Yang, Bo Zhang and Haiwen Zhang, A Nyström method for scattering by a two-layered medium with a rough boundary, *Journal of Computational and Applied Mathematics* 475 (2026), 116994.
- 6 Long Li, Jiansheng Yang, Bo Zhang and Haiwen Zhang, Uniform far-field asymptotics of the two-layered Green function in 2D and application to wave scattering in a two-layered medium, *SIAM Journal on Mathematical Analysis* 56 (2024), pp. 4143-4184.
- 5 Long Li, Jiansheng Yang, Bo Zhang and Haiwen Zhang, Direct imaging methods for reconstructing locally rough interfaces from phaseless total-field data or phased far-field data, *SIAM Journal on Imaging Sciences* 17 (2024), pp. 188-224.
- 4 Long Li, Guanghui Hu and Jiansheng Yang, Piecewise-analytic interfaces with weakly singular points of arbitrary order always scatter, *Journal of Functional Analysis* 284 (2023), 109800 (31pp).
- 3 Long Li, Jiansheng Yang, Bo Zhang and Haiwen Zhang, Imaging of buried obstacles in a two-layered medium with phaseless far-field data, *Inverse Problems* 37 (2021), 055004 (26pp).
- 2 Guanghui Hu, Long Li and Jun Zou, Unique determination of a penetrable scatterer of rectangular type for inverse Maxwell equations by a single incoming wave, *Inverse Problems* 35 (2019), 035006 (17pp).
- 1 Long Li, Guanghui Hu and Jiansheng Yang, Interface with weakly singular points always scatter, *Inverse Problems* 34 (2018), 075002 (13pp).

Selected Grants and Fellowships

- awarded 2026 Subwavelength resonances in coupled acoustic-elastic systems, FWF Erwin Schrödinger Fellowship, Austrian Science Fund (FWF), Grant No. J 5102-N. (Host: ETH Zürich; return phase: RICAM, Austrian Academy of Sciences).
- 2022.11–2024.01 Uniqueness and efficient reconstruction algorithm for inverse scattering problems in a two-layered medium, Principal Investigator, Grant No. 2022M720158a, China Postdoctoral Science Foundation (first class).

Teaching Experience

- 2020.03–07 **Teaching Assistant**, Advanced Mathematics B, Peking University.
- 2018.09–2019.01 **Teaching Assistant**, Advanced Mathematics B, Peking University.
- 2017.09–2018.01 **Teaching Assistant**, Design and Analysis of Algorithms, Peking University.
- 2017.03–07 **Teaching Assistant**, Advanced Mathematics C, Peking University.

Invited Talks

- 2026.08 High-Contrast Transmission and Fabry–Pérot-Type Resonances, Isaac Newton Institute for Mathematical Sciences, Cambridge, UK.
- 2025.09 Long-Time dynamics of acoustic resonators and applications in inverse problems, the 2025 ÖMG–DMV Annual Meeting, Linz, Austria.
- 2025.07 Uniform space and time behavior for acoustic resonators, the 3rd HKSIAM Biennial Conference, HongKong, China.
- 2025.02 Uniform resolvent estimates for subwavelength resonators, the 3rd Conference on Mathematics of Wave Phenomena, Karlsruhe, Germany.
- 2023.10 Piecewise-analytic interfaces with weakly singular points of arbitrary order always scatter, the 21st Annual Conference of the China Society for Industrial and Applied Mathematics, Kunming, China.
- 2023.07 Direct imaging methods for reconstructing locally rough interfaces from phaseless total-field data, the 13th Chinese Conference on Computational Mathematics, NanJing, China.
- 2021.10 Imaging of buried obstacles in a two-layered medium with phaseless far-field data, the 19th Annual Conference of the China Society for Industrial and Applied Mathematics, Hefei, China.

Honors and Awards

- 2021 **Outstanding Graduate of Peking University.**
- 2016-2021 **Peking University President Scholarship for Doctoral Students**, awarded for five consecutive academic years.